Local Projections or VARs: Evidence from Monte-Carlo Simulations

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SPS Seminar
Summer Semester 26

July 1, 2026

Motivation

- VARs and Local Projection used for Impulse Response Analysis, think Unemployment and Interest Rate, Oil Prices and Inflation, ...
- finite sample comparison (VAR(p) vs. LP(p))
  1. In finite samples, bias-variance puzzle
  2. VARs more efficient, LPs more robust to misspecification
- In Population both estimate the same Impulse Response
- Ludwig identity shows that equal lag comparison is inherently flawed

(BL:) Where does a fixed LP(4) sit on the VAR complexity frontier and does the conventional equal-lag ranking sruvive?
(EX:) Does any genuine LP advantage emerge once misspecification is introduced?

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Agenda

1. Theory
2. Study Design
3. Results
4. Discussion
5. Conclusion

VARs

$$Y_t = \nu + \mathbf{A}\,Y_{t-1} + U_t \xRightarrow[restr.K]{A1} y_t = \mu + \sum_{\ell=0}^{\infty} \Psi_\ell\, u_{t-\ell}, \qquad \Psi_\ell := [\mathbf{A}^\ell] \tag{1}$$

Identification: As $\sum_u = \mathbb{E}[u_t u_t'] \neq I_K$, Define $u_t = B\varepsilon_t$ where $\varepsilon_t \sim (0, I_k)$

$$y_{t+h} = \mu + \sum_{\ell=0}^{\infty} \Theta_\ell\, \varepsilon_{t+h-\ell} \tag{2}$$

Suffer from

- LS bias compunds as horizon grows Kilian 1998
- near-unit root behavior
- misspecification through MA dynamics or lag order

LPs

$$y_{t+h} = \alpha_h + \beta_h \, s_t + \boldsymbol{\gamma}_h' \, \mathbf{x}_t + \xi_{t+h}^{(h)}, \qquad h = 0, 1, \ldots, H, \tag{3}$$

Equivalence and Ludwig identity

DGPs

Method

Baseline Results - Complexity Frontier

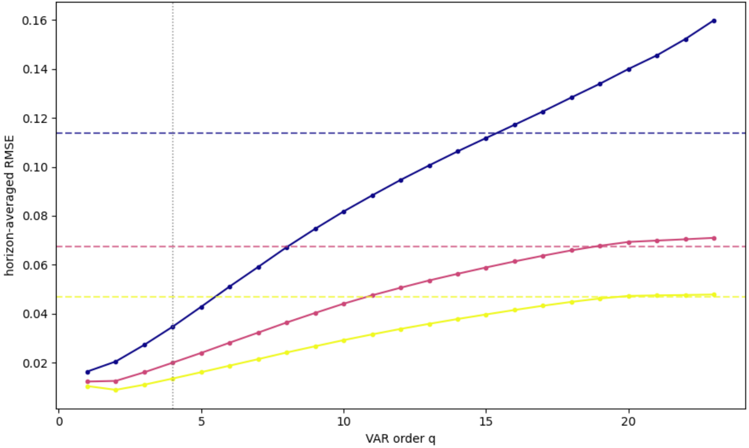


Figure 1: Complexity Frontier $\rho = 0.5$

Complexity Frontier - varying ρ

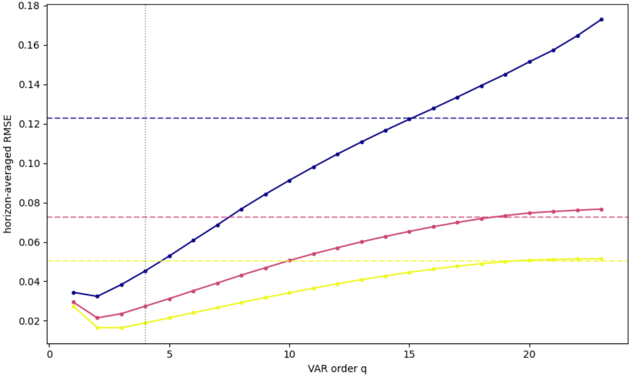


Figure 2: Complexity Frontier $\rho = 0.7$

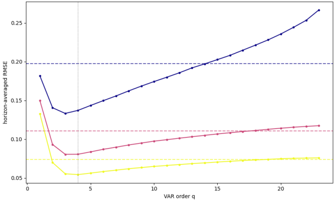
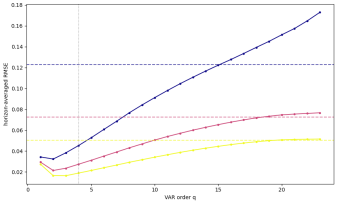


Figure 3: Complexity Frontier $\rho = 0.95$

Complexity Frontier - varying ρ



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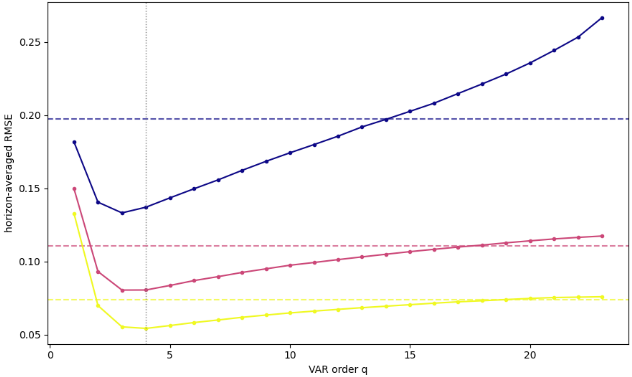


Figure 3: Complexity Frontier $\rho = 0.95$

Bias

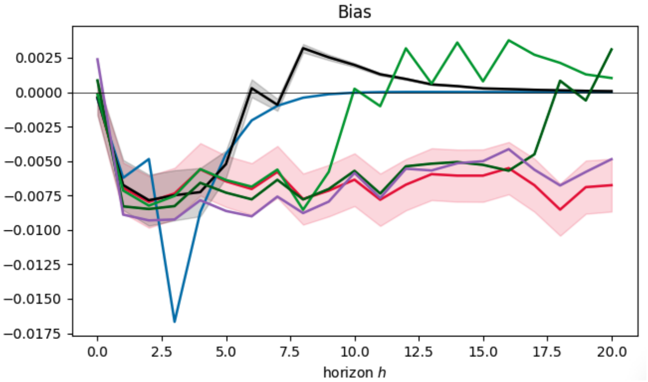


Figure 4: Bias at $\rho = 0.5$

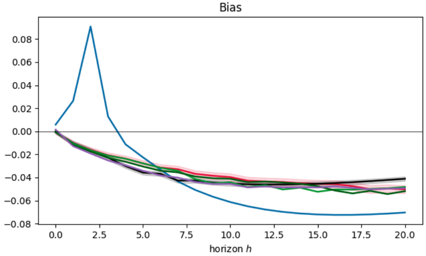


Figure 5: Bias at $\rho = 0.95$

Bias

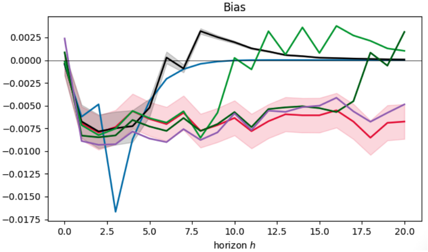


Figure 4: Bias at $\rho = 0.5$

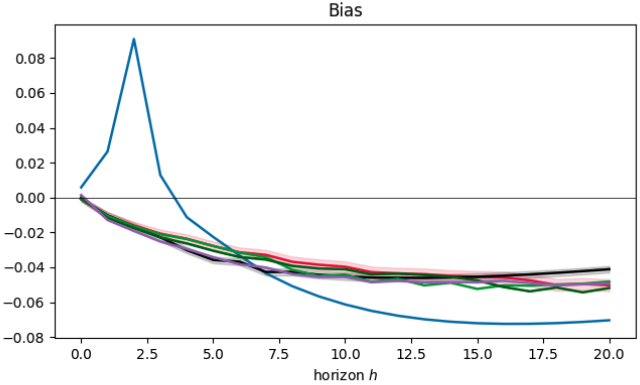


Figure 5: Bias at $\rho = 0.95$

Variance

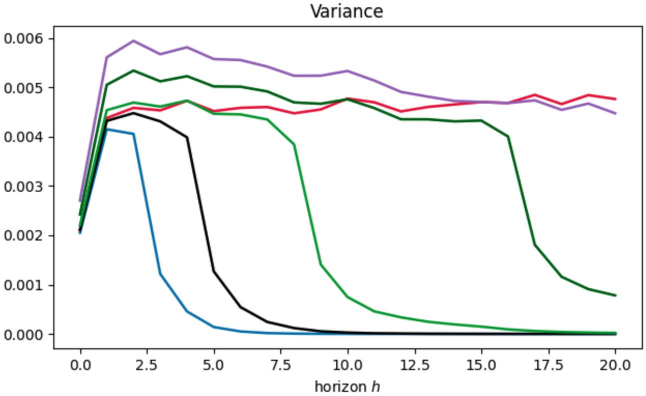


Figure 6: Variance at $\rho = 0.5$

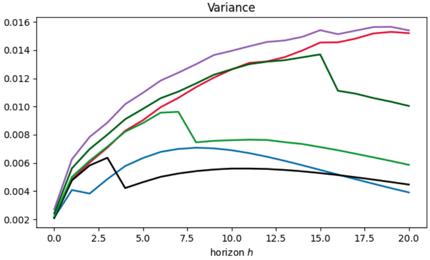


Figure 7: Variance at $\rho = 0.95$

Variance

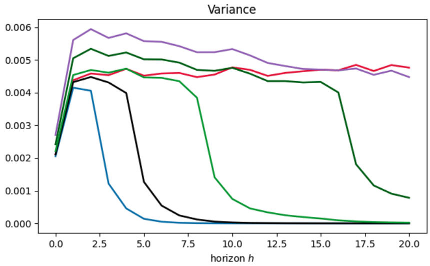


Figure 6: Variance at $\rho = 0.5$

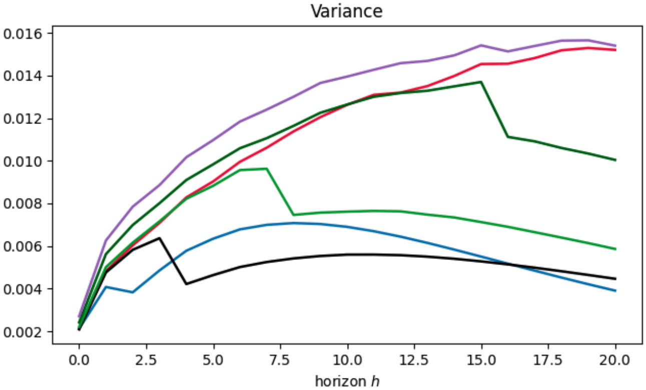


Figure 7: Variance at $\rho = 0.95$

Coverage

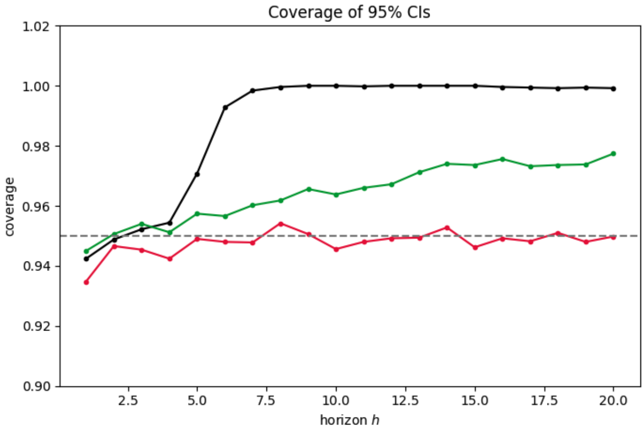


Figure 8: Coverage at $\rho = 0.5$

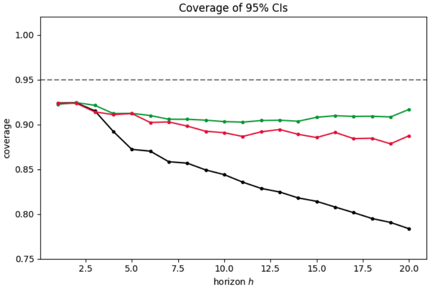
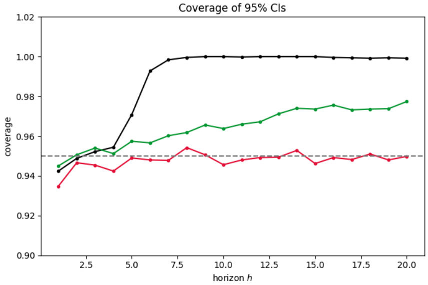


Figure 9: Coverage at $\rho = 0.95$

Coverage



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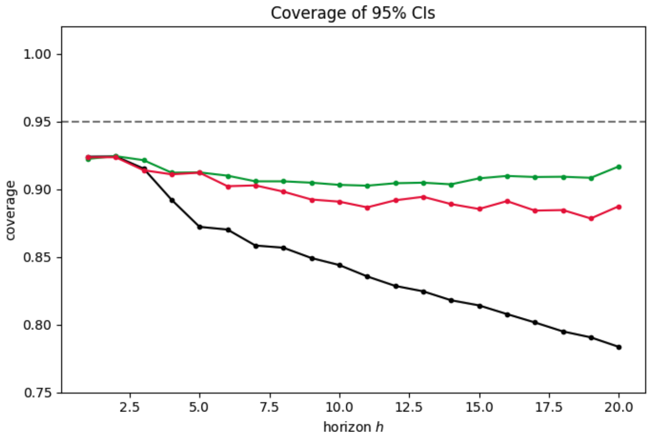

Figure 9: Coverage at $\rho = 0.95$

Introducing Dynamic Misspecification

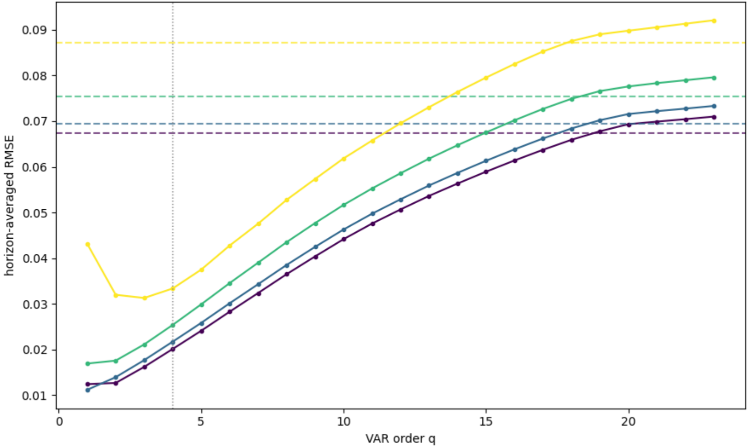

Figure 10: Complexity Frontier $\rho = 0.5$

Dynamic Misspecification - varying ρ

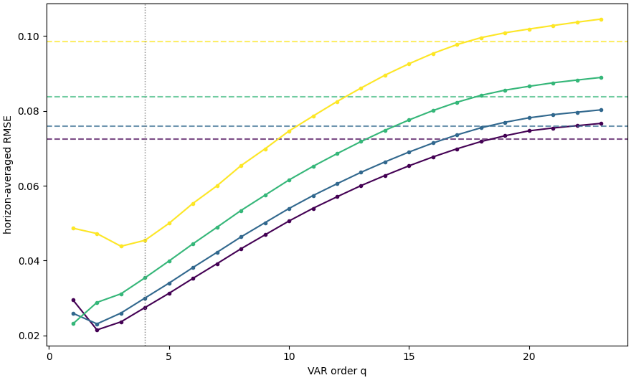

Figure 11: Complexity Frontier $\rho = 0.7$

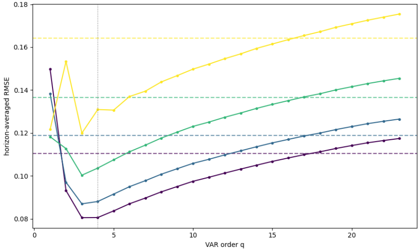

Figure 12: Complexity Frontier $\rho = 0.95$

Dynamic Misspecification - varying ρ

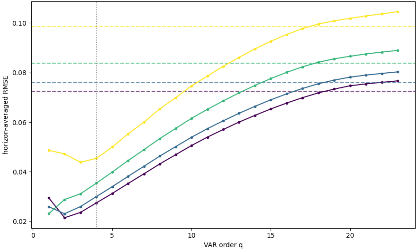

Figure 11: Complexity Frontier $\rho = 0.7$

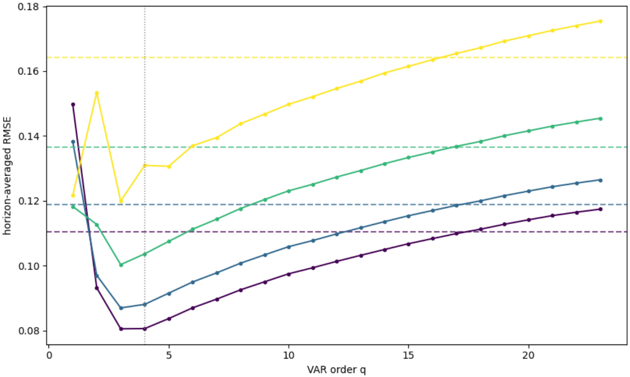

Figure 12: Complexity Frontier $\rho = 0.95$

Bias - Correct vs. Misspecification - $\tau = 0.6$

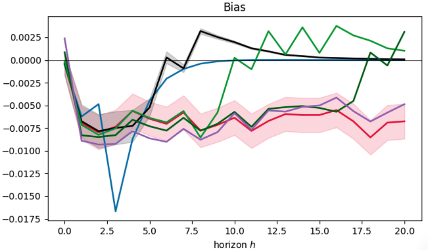

Figure 4: Baseline at $\rho = 0.5$

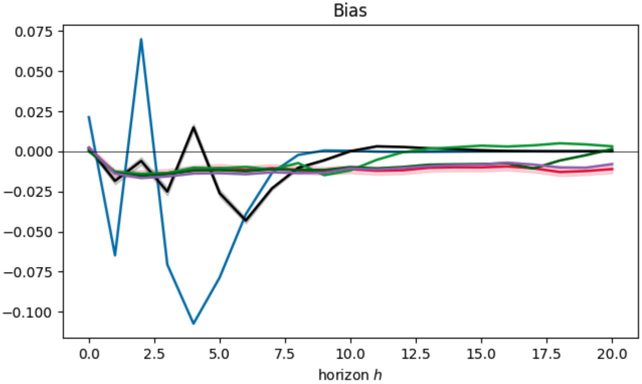

Figure 13: Dyn. Misspecification at $\tau = 0.6$

Variance - Correct vs. Misspecification - $\tau = 0.6$

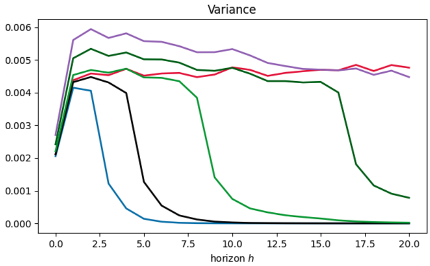

Figure 6: Baseline at $\rho = 0.5$

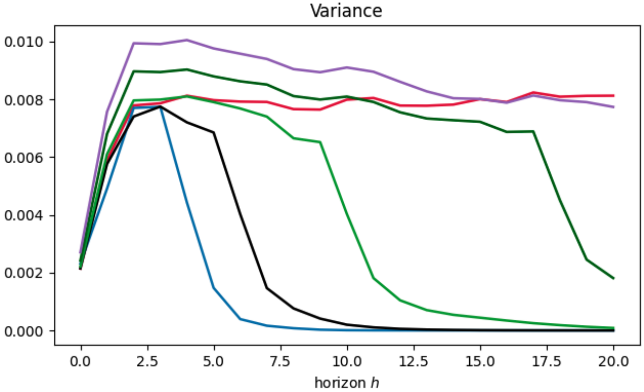

Figure 14: Dyn. Misspecification at $\tau = 0.6$

Coverage - Correct vs. Misspecification - $\tau = 0.6$

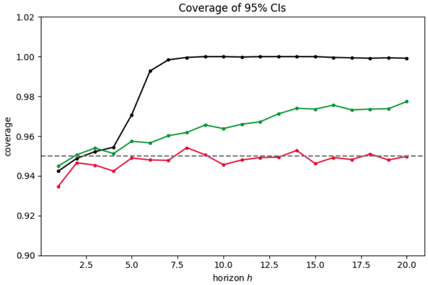

Figure 8: Baseline at $\rho = 0.5$

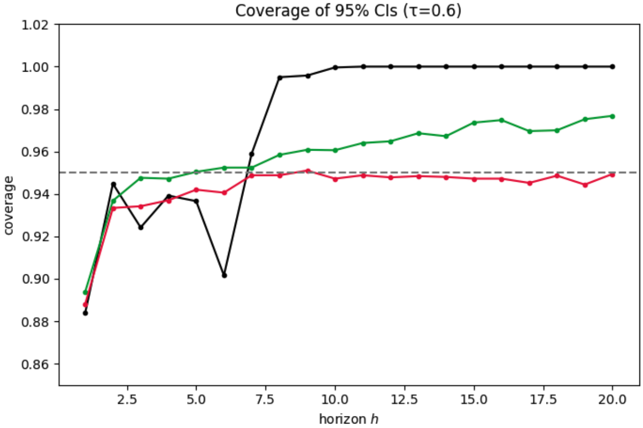

Figure 15: Dyn. Misspecification at $\tau = 0.6$

Introducing Functional Misspecification

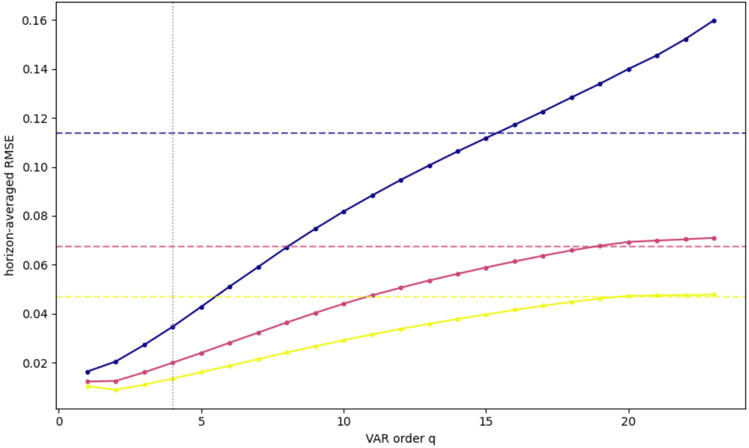

Figure 16:

Variance - Correct vs. Misspecification - $\rho = 0.5$

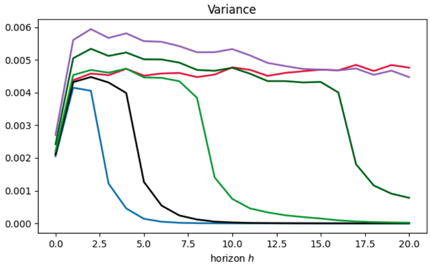

Figure 6: Baseline at $\rho = 0.5$

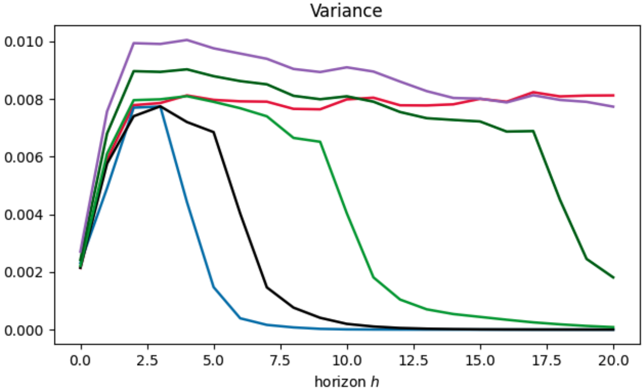

Figure 14: Dyn. Misspecification at $\gamma = 0.6$

Coverage - Correct vs. Misspecification - $\rho = 0.5$

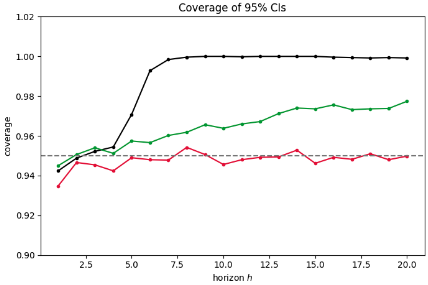

Figure 8: Baseline

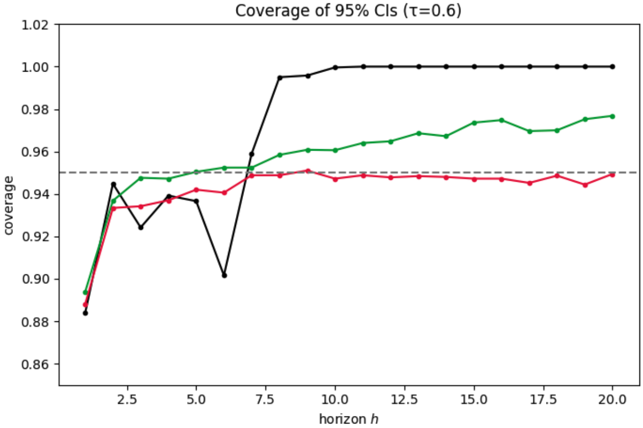

Figure 15: Fct. Misspecification at $\gamma = 0.6$

Discussion of Results

- LP(4) $\approx$ VAR(15)
- As T decreases, LP(4) is equal to a less parsimonous VAR $\rightarrow$ relatively better
- As ρ increases, LP(4) is equal to a less parsimonous VAR $\rightarrow$ relatively better
- As τ increases, LP(4) is equal to a less parsimonous VAR $\rightarrow$ relatively better
- At equal lags, VAR(p) is preferable choice if applicable, except for coverage
- higher order VARs closes coverage gap, at cost of bias and variance

Conclusion

1. Ludwig identity largely consistent with MC findings
2.

Thank you for your Attention! Questions?